Presentation Script

Challenges & Limitations — Motion-Sensing Living Room Game | Read-aloud script, one section per slide

Slide 1 — Title Slide

Hi everyone, thanks for having us. Today we want to walk you through the biggest challenges we ran into while building our motion-sensing living room game, and how we tackled each one — including the limitation we haven't fully solved yet. Let's get into it.

Slide 2 — What We're Building

Images that: we're building a motion-sensing game for the living room. Instead of a controller, the camera or your phone tracks your skeleton in real time and turns your movement into gameplay. And unlike a lot of motion-tracking projects out there, we built this for casual, everyday use in a shared space — not for one person wearing a VR headset alone. Keeping that goal in mind is actually the key to understanding our first challenge.

Slide 3 — Challenge 1 — Wrong Platform, Wrong Audience

Our first challenge was actually about picking the wrong tool for the job. We originally built on Apple's ARKit, which gives really precise 3D skeletal data — but it's designed for virtual reality users, and it only works on newer Apple devices. That's a great fit for VR, but not for us: our goal is a casual living-room game that as many families as possible can play. So even though VR tracking and our tracking overlap technically, they serve completely different audiences. We made the call to move to cross-platform 2D skeletal data instead. We lose some precision, but we gain compatibility with most everyday smartphones, which matters a lot more for reaching a broad audience than chasing the most advanced technology.

Slide 4 — Challenge 2 — The Training-Data Bottleneck

Our second challenge was about data, not code. The core of our product is recognizing what action someone is performing from video. To get that reliable enough for a real product, you typically need a massive labeled dataset — and building that from scratch just wasn't realistic for a team our size and timeline. So we made a strategic call: instead of training a model from zero, we adopted pre-built solutions from major tech companies that can already predict skeletal data streams directly from video. This let us skip years of data collection and get to production-level reliability much faster, by standing on the shoulders of research that already exists.

Slide 5 — Challenge 3 — One Living Room, Many Players

Next challenge: our game only supported one player on one device. But that doesn't match how living rooms actually get used — it's rarely just one person, it's family game night, or friends hanging out. So we had to rethink the architecture to support multiple users at the same time. Our solution was to build a high-performance gateway: a layer that lets several devices — phones, cameras — connect at once, with each one streaming a different player's skeletal data into the same shared game session in real time. That turned our single-player prototype into something that actually fits how a living room works.

Slide 6 — An Honest Limitation We Still Have

We also want to be upfront about a limitation we haven't fully solved. The gateway solves the multi-device connection problem, but each individual camera or phone can still only track one person's skeleton at a time. So in practice, right now, we need roughly one device per player — the gateway scales the number of players in the room, but it doesn't turn a single camera into a system that tracks several people at once. That's a real constraint on the current version, and it's something we're transparent about rather than glossing over.

Slide 7 — Challenge 4 — Beyond Full-Body Tracking

Our last challenge is really more of an opportunity we're just starting to explore. Right now, our model only captures full-body data. But full-body isn't the only thing worth tracking — a camera could just as easily follow a waved game prop, like a toy sword, or focus in tightly on finger movement for something more precise. The good news is that our new gateway was designed with this in mind: it can support different skeletal models, not just full-body, and route that data into whatever kind of game needs it. So this is less a limitation and more the next direction we want to grow in.

Slide 8 — Putting It All Together

So to pull that all together: four challenges, four deliberate decisions. Wrong audience for our data format — we switched to cross-platform 2D tracking. Not enough data to train our own model — we adopted an existing pre-built solution. Only single-user support — we built a gateway for multiple devices. And full-body-only tracking — we designed the gateway to be flexible for other kinds of skeletal data down the line. None of these were the flashiest technical choices, but each one moved us closer to a product real people in a real living room could actually use.

Slide 9 — Key Takeaways

Before we wrap up, here's what we'd want other teams to take away from our experience. First, design for your real audience — the most advanced technology isn't automatically the right choice; ARKit was more powerful, but it was solving the wrong problem for us. Second, don't reinvent what already exists — using a proven pre-built model saved us from a data-collection effort we simply didn't have the resources for. And third, build flexible infrastructure early — the gateway we originally built just to support multiple users ended up being exactly what let us start exploring new kinds of tracking, too.

Slide 10 — Thank You / Questions

That covers the four biggest challenges we faced — getting the platform and audience right, solving the training-data bottleneck, adding multi-user support through our gateway, and starting to expand beyond full-body tracking. Thanks so much for listening — we'd love to hear your questions.